Problem 1. Completely read the Homework Guidelines that are posted on Canvas.

Problem 2. Let x be an even integer and let n be a natural number. Prove that x^n is an even integer.

Problem 3. Let $n \in \mathbb{N}$. (In other words, n is a natural number.)

1. Make at least two conjectures about factorizations/primeness of $3^n - 1$.
2. Make at least two conjectures about factorizations/primeness of $3^n - 2^n$.

For each of these, you can make a table like in the example we did on the first day of class to guide your conjectures. You do not need to prove your conjectures.

Note: A conjecture should be something you suspect is true based on a reasonable amount of evidence, but you don't know for sure. Technically, "$3^n - 1$ is prime for every value of n" might be a conjecture, but it is not a very good one, because $3^2 - 1$ is not prime, so the conjecture is clearly false. Also, technically "$3^n - 1$ is an integer" might be a conjecture, but not a good one since it is obviously true and does not address primeness.

Problem 4. Let P be the statement "I went hiking last weekend", let Q be the statement "Greek olives are from Italy," and R be the statement "Birds are not real." Write the following logical expressions in full English sentences.

1. $\neg(P \vee Q) \wedge R$.
2. $(Q \vee \neg R) \vee (\neg Q \wedge P)$.
3. $\neg[P \wedge (\neg R \vee \neg Q)] \vee \neg Q$.

Problem 5. For the following sentences, assign P, Q, etc. to each statement and write these as logical expressions. (In other words, do the opposite of the previous problem.)

1. Either Kathy has been fired or she has quit her job, and she's moving to Argentina and she's never coming back.
2. I can't type very fast, but this document will be finished eventually.
3. Snoopy is either hungry and he is not tired, or he is not hungry and he is angry.

Problem 6. Make truth tables for the following formulas.

1. $\neg[P \wedge (Q \vee \neg P)]$.
2. $(P \wedge Q) \vee (\neg P \wedge \neg Q)$.
3. $(P \vee Q) \wedge (P \vee \neg R)$.

Problem 7. Find a formula using the connectives $\wedge$, $\vee$, and $\neg$ that has the following truth table.

P	Q	???
T	T	F
T	F	T
F	T	T
F	F	F

Problem 8. Use truth tables to prove the following.

1. $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$. (One of DeMorgan's laws)
2. $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$. (One of the distributive laws)

Problem 9. For each of the following, use truth tables to determine whether the expression is a tautology, a contradiction, or neither.

1. $\neg Q \vee (P \to Q)$.
2. $(Q \wedge P) \wedge (P \to \neg Q)$.
3. $Q \vee (P \wedge \neg Q)$.
4. $\neg Q \to (P \wedge \neg P)$.

Problem 10. Let P be the statement "He is van Gogh" and Q be the statement "He cut his ear." Write the conditional statement $P \to Q$ in English using the following terms.

1. Using the words "if" and "then."
2. Using the word "implies."
3. Using the phrase "only if."
4. Using the phrase "is necessary for."
5. Using the phrase "is sufficient for."

Problem 11 (Practice.)**.** For extra practice, I recommend completing the progress checks in sections 1.1, 2.1 and 2.2 of the textbook. Do not hand these in. Just think about the problems on your own. You can look up solutions in the back of the book.